

$$(1.1) \quad X = \min \{X_1, X_2, X_3\}$$

$$F_X(x) = P[X < x]$$

$$= P[\min \{X_1, X_2, X_3\} < x]$$

$$= P[X_1 < x] + P[X_2 < x] + P[X_3 < x]$$

$$= \sum_{i=1}^3 P[X_i < x]$$

$$= \sum_{i=1}^3 F_{X_i}(x)$$

$$= \sum_{i=1}^3 -\frac{1}{\lambda} \ln(u)$$

$$(1.2) \quad f_Y(y) = \frac{3}{8} y^2 e^{-y^3/8}$$

$$F(y) = \int_0^y \frac{3}{8} t^2 e^{-t^3/8} dt$$

$$u = \frac{t^3}{8} \quad \frac{du}{dt} = \frac{3t^2}{8} \quad dt = \frac{8}{3t^2} du$$

$$F(y) = \int_0^{y^3/8} e^{-u} du = -e^{-u} \Big|_0^{y^3/8} = -e^{-y^3/8} + e^0 = 1 - e^{-y^3/8}$$

$$F(y) = u \Rightarrow u = 1 - e^{-y^3/8}$$

$$e^{-y^3/8} = 1 - u$$

$$-y^3/8 = \ln(1-u)$$

$$y^3 = -8 \ln(1-u)$$

$$y = (-8 \ln(1-u))^{1/3} = F^{-1}(y)$$

$$\textcircled{2} P(X=k) = \frac{e^{-10} 10^k}{k!}$$

$$P[X \leq k] = \sum_1^k \frac{e^{-10} 10^k}{k!} \leftarrow$$

① draw U

② value of U is $P[X \leq k]$

③ so figure out what k is, given U

↳ you can just compute the summation

$$\textcircled{3.1} F_X(x) = \int_0^{\infty} 2x^y e^{-2y} dy$$

$$x^y = e^{y \ln(x)}$$

$$F(x) = \int_0^{\infty} 2 e^{y(\ln(x)-2)} dy \quad c = \ln(x) - 2 \leftarrow \text{since } 0 < x \leq 1 \quad c < 0$$

$$F(x) = \int_0^{\infty} 2 e^{cy} dy = \int_0^{\infty} e^{cy} dy = \frac{1}{c} e^{cy} \Big|_0^{\infty} = \underbrace{\frac{1}{c} e^{-\infty}}_{\text{approaches zero}} - \frac{1}{c} e^0 = -\frac{1}{c}$$

$$F(x) = -\frac{1}{c} = -\frac{1}{\ln(x)-2}$$

$$u = -\frac{1}{\ln(x)-2}$$

$$\ln(x) = -\frac{1}{u} + 2$$

$$x = e^{-\frac{1}{u} + 2}$$

$$F^{-1}(x) = e^{-\frac{1}{u} + 2}$$

$$(3.2) f_Y(y) = 2e^{-2y}$$

$$F_Y(y) = \int 2e^{-2y} dy = -e^{-2y}$$

$$u = e^{-2y}$$

$$\ln(-u) = -2y$$

$$y = -\frac{1}{2} \ln(-u)$$

① simulate y

② simulate x

$$(5) X = (X, Y) \quad \leftarrow \text{because are of circles } \pi$$

$$f(x, y) = \begin{cases} \frac{1}{\pi} & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x, y) = \frac{f(x, y)}{f(y)} f(y) = f_{X|Y}(x, y) \cdot f(y)$$

$$f_{X|Y}(x, y) = \frac{f(x, y)}{f(y)}$$